

# DSA 210 Term Project Final Report

## The Impact of Travel Distance and Weather Conditions on UEFA Champions League Away Performances

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### 1. Introduction and Motivation

Football match outcomes are affected by many different factors. Team quality, tactical decisions, squad depth, injuries, player form, home advantage, travel burden, and environmental conditions can all influence how a team performs. Among these factors, away-team performance is especially interesting because away teams often face additional disadvantages compared to home teams. They travel to another city or country, play in an unfamiliar stadium, and may face different weather or environmental conditions.

The motivation of this project is to investigate whether two external factors, **travel distance** and **weather conditions**, are associated with away-team performance in UEFA Champions League matches. The UEFA Champions League is a suitable competition for this question because it includes clubs from many different countries, and teams often travel long distances across Europe. In addition, matches are played in different cities and climates, which creates natural variation in temperature, precipitation, and wind conditions.

My initial expectation was that longer travel distances and more difficult weather conditions could negatively affect away-team performance. For example, I expected that teams traveling longer distances might perform worse because of fatigue, travel burden, and adaptation issues. Similarly, I expected that rain, wind, or extreme temperatures could make matches more difficult and possibly reduce away-team success. However, one of the main goals of this project was not only to confirm this intuition, but to test it using data.

The central research question of this project is:

**Are travel distance and match-day weather conditions associated with away-team performance in UEFA Champions League matches?**

To answer this question, I created a complete data science pipeline. The project starts from a public football dataset, filters UEFA Champions League matches, cleans and prepares the data, creates away-performance variables, enriches the dataset with stadium location information and

historical weather data, estimates away-team travel distance, performs exploratory data analysis, applies hypothesis testing, and finally uses machine learning models to evaluate whether away wins can be predicted from weather and travel-related variables.

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## 2. Dataset and Data Sources

The main dataset used in this project is a football match-level dataset. The central file is `games.csv`, which includes information about football matches such as competition ID, season, round, match date, home club, away club, goals, stadium, attendance, and other match-related variables.

For this project, I focused specifically on UEFA Champions League matches. I filtered the dataset using the competition ID corresponding to the Champions League. After filtering, the dataset contained UEFA Champions League match records that could be used for the analysis.

The main variables used from the football dataset were:

- match date
- home club name
- away club name
- home club goals
- away club goals
- stadium name
- attendance
- competition ID

In addition to the original football dataset, I created and used several enriched datasets:

- `stadium_reference_table_filled.csv`
- `match_weather_data.csv`
- `final_enriched_dataset.csv`
- `ml_enriched_dataset.csv`

The `stadium_reference_table_filled.csv` file contains stadium names, cities, countries, latitudes, and longitudes. This file was necessary because the original football dataset included stadium names, but did not directly provide geographic coordinates for all stadiums.

The `match_weather_data.csv` file contains match-day weather variables retrieved using match dates and stadium coordinates. The weather data was retrieved from the Open-Meteo historical weather API.

The `final_enriched_dataset.csv` file combines the cleaned football dataset with stadium location data and weather variables.

The `ml_enriched_dataset.csv` file further includes estimated travel distance for away teams and was used in the machine learning stage.

The weather variables used in the project were:

- `temperature_2m_mean`
- `precipitation_sum`
- `windspeed_10m_max`

The travel-related variable used in the project was:

- `travel_distance_km`

This variable represents the estimated geodesic distance between the away club's representative home stadium and the match stadium.

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### 3. Data Cleaning and Preprocessing

The first major step was filtering the football dataset to UEFA Champions League matches only. Since the raw dataset contained matches from many different competitions, I used the `competition_id` column to select the relevant Champions League records.

After filtering the data, I selected the variables needed for the analysis:

- `date`
- `home_club_name`
- `away_club_name`
- `home_club_goals`
- `away_club_goals`
- `stadium`
- `attendance`

The `date` column was converted into a proper datetime format. This was important because the date variable was later used to retrieve historical weather data for each match day.

I then checked missing values in the selected columns. The most important columns for the project were match date, club names, goals, and stadium names. Rows with missing values in these core variables were removed to create a clean match-level dataset. Some missing values existed in the attendance column, but attendance was not used as a core variable in the hypothesis tests or machine learning models. Therefore, missing attendance values did not create a major problem for the main analysis.

The purpose of this cleaning stage was to create a reliable dataset in which each row represented one UEFA Champions League match with the necessary information for later enrichment and analysis.

After cleaning, the dataset was ready for feature engineering and external data enrichment.

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## 4. Feature Engineering

After cleaning the dataset, I created new variables to describe away-team performance more directly. The original dataset included home goals and away goals, but I needed variables that directly represented the away team's result.

The first variable was:

```
away_win
```

This variable equals 1 if the away team scored more goals than the home team, and 0 otherwise. It was calculated as:

```
away_win = 1 if away_club_goals > home_club_goals, otherwise 0
```

The second variable was:

```
away_not_lose
```

This variable equals 1 if the away team either won or drew the match, and 0 if the away team lost. It was calculated as:

```
away_not_lose = 1 if away_club_goals >= home_club_goals, otherwise 0
```

The third variable was:

```
away_goal_difference
```

This variable was calculated as:

```
away_goal_difference = away_club_goals - home_club_goals
```

This is a useful numerical performance measure. If the value is positive, the away team won. If it is zero, the match ended in a draw. If it is negative, the away team lost.

These engineered variables were important because they allowed the project to analyze away-team performance in different ways. `away_goal_difference` was especially useful for hypothesis testing because it is numerical and contains more information than a simple win/loss

variable. `away_win` was useful for machine learning because it created a binary classification target.

The exploratory statistics showed that away teams won less often than they failed to win, which is consistent with the idea of home advantage. The average away goal difference was also negative, showing that away teams generally performed slightly worse than home teams.

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## 5. Weather Data Enrichment

A major part of the project was enriching the football dataset with match-day weather data. The original football dataset did not include weather information, so I created an additional enrichment pipeline.

First, I needed stadium location information. I created a stadium reference table containing:

- stadium name
- city
- country
- latitude
- longitude

This table was then merged with the match-level dataset using the stadium name. After this merge, each match had location information for the stadium where it was played.

Then, I used the match date together with the stadium latitude and longitude to retrieve historical weather data from the Open-Meteo historical weather API.

The weather variables collected were:

```
temperature_2m_mean  
precipitation_sum  
windspeed_10m_max
```

These variables represent average daily temperature, total daily precipitation, and maximum daily wind speed for the match location and match date.

Before applying the weather retrieval process to the full dataset, I tested the API call on a small sample. This was important because API calls can fail due to formatting problems, connection issues, or invalid coordinates. After confirming that the sample weather retrieval worked correctly, I applied the process to the full dataset.

The full weather enrichment process successfully retrieved weather data for almost all matches. A small number of missing weather values remained. Since the number of missing weather

records was very small compared with the full dataset, I removed those rows before performing hypothesis testing and machine learning.

This enrichment step was important because it allowed the project to move beyond the original football dataset and include external environmental variables.

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## 6. Travel Distance Estimation

The second major enrichment step was estimating travel distance for away teams. Exact travel itineraries were not available, so I approximated away-team travel burden using geodesic distance.

The main idea was to calculate the distance between:

1. the away club's representative home stadium
2. the stadium where the match was played

To do this, I first identified each club's representative home stadium. I did this by looking at matches where the club appeared as the home team and selecting the stadium most frequently associated with that club.

Then, I matched each representative home stadium with the stadium reference table to obtain home latitude and longitude.

For each match, I merged the away club's home stadium coordinates into the dataset. Then, I calculated the geodesic distance between:

`away club home coordinates`

and

`match stadium coordinates`

The resulting variable was:

`travel_distance_km`

This variable represents the estimated travel distance of the away team in kilometers.

This approach has limitations because it does not use actual flight routes or real travel itineraries. However, it provides a reasonable approximation of travel burden using the available data.

The travel distance variable showed substantial variation across matches. Some away teams traveled relatively short distances, while others had much longer trips. The maximum values were high, which is reasonable for an international competition such as the UEFA Champions League.

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## 7. Exploratory Data Analysis

The exploratory data analysis focused on understanding the structure of away-team performance before formal statistical testing.

The away win distribution showed that away wins were less common than non-away-wins. Away teams won approximately one-third of the matches. This means that the target variable `away_win` was somewhat imbalanced, with non-away-wins being the majority class.

The average away goal difference was negative. This means that, on average, away teams scored fewer goals than home teams. This result is consistent with the general idea of home advantage in football.

The distribution of `away_goal_difference` showed that many matches were close, but negative goal differences were common. This again suggested that away teams generally faced a disadvantage.

I also compared average home goals and away goals. The average number of home goals was higher than the average number of away goals. This supported the same pattern: home teams generally performed better than away teams in the dataset.

After enriching the dataset, I explored weather and distance variables. Weather variables such as temperature, precipitation, and wind speed showed variation across matches. Travel distance also varied substantially.

However, visual exploration did not suggest a strong relationship between these external factors and away-team performance. This was an important early observation. Although it was intuitive to expect that worse weather or longer travel might harm away teams, the descriptive patterns did not show a clear strong effect.

This exploratory finding motivated the next stage: formal hypothesis testing.

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## 8. Hypothesis Testing

The hypothesis testing stage examined whether weather conditions and travel distance were statistically associated with away-team performance. The main performance variable used in this stage was:

away\_goal\_difference

This variable was used because it provides a continuous measure of performance and is more informative than a binary win/loss outcome.

## 8.1 Temperature and Away Goal Difference

The first hypothesis test examined the relationship between mean temperature and away goal difference.

The correlation coefficient between `temperature_2m_mean` and `away_goal_difference` was approximately:

-0.0028

The p-value was approximately:

0.9046

This correlation is extremely close to zero, and the p-value is very high. Therefore, there was no statistically significant linear relationship between temperature and away goal difference.

This result suggests that average match-day temperature alone did not explain away-team performance in this dataset.

## 8.2 Rainy vs Non-Rainy Matches

The second test examined precipitation. I divided matches into two groups:

```
rainy matches: precipitation_sum > 0  
non-rainy matches: precipitation_sum = 0
```

Then, I compared away goal difference between rainy and non-rainy matches using a t-test.

The t-statistic was approximately:

-0.359

The p-value was approximately:

0.719

The average away goal difference was slightly worse in rainy matches than in non-rainy matches, but the difference was very small and not statistically significant.

This means that the data did not provide strong evidence that rainy match conditions significantly affected away-team goal difference.



### 8.3 High-Wind vs Low-Wind Matches

The third weather-based test examined wind speed. I divided matches into high-wind and low-wind groups using the median value of `windspeed_10m_max`.

The t-statistic was approximately:

0.716

The p-value was approximately:

0.474

The average away goal difference differed slightly between the two groups, but the difference was not statistically significant.

This means that wind speed did not show a statistically significant relationship with away-team goal difference in the dataset.

### 8.4 Travel Distance and Away Goal Difference

The next test examined the relationship between estimated travel distance and away goal difference.

The correlation coefficient between `travel_distance_km` and `away_goal_difference` was approximately:

-0.0145

The p-value was approximately:

0.534

This correlation is very close to zero, and the p-value is high. Therefore, estimated travel distance did not show a statistically significant linear relationship with away goal difference.

This result was somewhat surprising because my initial expectation was that longer travel distance might negatively affect away-team performance. However, the data did not provide strong statistical support for that expectation.

### 8.5 Long-Distance vs Short-Distance Matches

Finally, I divided matches into short-distance and long-distance groups using the median travel distance.

The average away goal difference in long-distance matches was approximately:

-0.402

The average away goal difference in short-distance matches was approximately:

-0.368

This suggests that long-distance matches had slightly worse average away goal difference. However, the t-test p-value was approximately:

0.732

This means the difference was not statistically significant.

Overall, the hypothesis testing results suggest that weather conditions and estimated travel distance did not show strong statistically significant associations with away-team performance in this dataset.

This was an important result because it challenged the initial intuition of the project. While travel and weather may matter in specific match contexts, their effects were not strong enough to appear as statistically significant patterns in the aggregated match-level data used here.

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## 9. Machine Learning Analysis

The machine learning stage investigated whether away-team wins could be predicted using weather and travel-related variables.

The target variable was:

`away_win`

This variable equals 1 when the away team wins and 0 otherwise.

The selected input features were:

- `temperature_2m_mean`
- `precipitation_sum`
- `windspeed_10m_max`
- `travel_distance_km`

Rows with missing values in these variables were removed before model training. After cleaning, the machine learning dataset contained 1819 rows.

The dataset was split into training and test sets using an 80-20 split. Stratified splitting was used because the target variable was imbalanced. This ensured that the class distribution was similar in both the training and test sets.

Five classification models were trained and evaluated:

1. Baseline Logistic Regression
2. Balanced Logistic Regression
3. Decision Tree Classifier
4. Random Forest Classifier
5. Balanced Random Forest Classifier

## 9.1 Baseline Logistic Regression

The baseline Logistic Regression model achieved an accuracy of approximately:

0.668

At first glance, this seemed like a reasonable result. However, the classification report and confusion matrix showed an important problem. The model almost always predicted the majority class, meaning it mostly predicted that the away team would not win.

The recall for the positive class, `away_win = 1`, was:

0.00

This means the model failed to correctly identify away wins.

This result showed that accuracy alone can be misleading in an imbalanced classification problem. The model looked acceptable based on raw accuracy, but it was not useful for detecting away wins.

## 9.2 Balanced Logistic Regression

To address the class imbalance problem, I trained a Logistic Regression model with class balancing.

The Balanced Logistic Regression model had lower accuracy, approximately:

0.478

However, it improved recall for away wins to approximately:

0.420

This means that the balanced model was much better at detecting the minority class than the baseline Logistic Regression model.

The trade-off was that the model also made more false positive predictions. In other words, it predicted more away wins, which helped it find actual away wins, but also increased mistakes.

This model demonstrated that class balancing can make the model more informative when the goal is to detect away wins rather than simply maximize overall accuracy.

### **9.3 Decision Tree Classifier**

The Decision Tree model was trained as a nonlinear benchmark. It achieved an accuracy of approximately:

0.637

However, its recall for away wins was only approximately:

0.050

This means that the Decision Tree still struggled to identify away wins. Although it was more flexible than Logistic Regression, it did not find strong predictive patterns in the selected weather and travel variables.

The feature importance results showed that precipitation and wind speed were used more strongly by the tree, while travel distance had little or no importance in the selected tree structure.

### **9.4 Random Forest Classifier**

The Random Forest model achieved an accuracy of approximately:

0.662

However, like the baseline Logistic Regression model, it mostly predicted the majority class. Its recall for away wins was:

0.00

This means that the standard Random Forest also failed to detect away wins effectively.

This result showed that using a more complex model does not automatically solve the problem if the selected features do not contain strong predictive information.

### **9.5 Balanced Random Forest Classifier**

Finally, I trained a Random Forest model with class balancing.

The Balanced Random Forest achieved an accuracy of approximately:

0.522

Its recall for away wins was approximately:

0.328

Its F1-score for away wins was approximately:

0.310

Compared with the standard Random Forest, the Balanced Random Forest was much better at detecting away wins. It did not achieve the highest raw accuracy, but it provided a more reasonable balance between overall performance and minority-class detection.

Among the tested models, the Balanced Random Forest was one of the most informative models because it did not completely ignore the minority class.

The machine learning results supported the earlier hypothesis-testing results. Weather and travel distance alone did not provide strong predictive power for away-team wins. However, balanced models showed that there is limited predictive signal, especially when the model is encouraged to pay attention to away wins.

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## 10. Results and Discussion

The overall results of the project show that away teams generally perform worse than home teams in UEFA Champions League matches. This was visible in the exploratory analysis through the lower away win rate and negative average away goal difference.

However, the main research variables of the project, weather conditions and estimated travel distance, did not show strong statistical or predictive relationships with away-team performance.

The hypothesis tests showed no statistically significant relationship between:

- temperature and away goal difference
- precipitation and away goal difference
- wind speed and away goal difference
- travel distance and away goal difference

The machine learning results were consistent with this conclusion. Models trained only on weather and travel-related variables struggled to predict away wins effectively. Some models achieved relatively high accuracy, but this was mostly because they predicted the majority class. In other words, they often predicted that the away team would not win, which was correct most of the time because away wins were less common.

The balanced models were more informative because they improved the detection of away wins. However, their performance was still modest. This suggests that weather and travel distance contain limited predictive information, but they are not strong standalone predictors of away-team success.

One of my most important observations from this project is that intuitive football assumptions do not always appear clearly in aggregated data. It is reasonable to expect that long travel or difficult weather might hurt away teams. However, in this dataset, these effects were either weak, inconsistent, nonlinear, or hidden behind stronger factors.

Football performance is likely influenced more strongly by variables such as:

- team quality
- player availability
- tactics
- recent form
- squad depth
- match importance
- competition stage
- home advantage
- injuries and suspensions

Therefore, the results do not necessarily mean that weather and travel never matter. Instead, they suggest that weather and travel distance alone are not enough to explain away-team performance in UEFA Champions League matches.

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## 11. Limitations

This project has several limitations.

First, travel distance was estimated using representative home stadium locations instead of exact travel routes. In reality, teams may not always travel from their home city. They may travel from a different location depending on their schedule, training camp, or previous match.

Second, the distance measure was based on geodesic distance, not actual flight or road distance. This provides a useful approximation, but it does not fully represent real travel burden.

Third, weather data was measured at the match-location and match-date level. The variables were daily weather measures, not exact kickoff-time conditions. Match-time weather could be different from daily averages or daily maximum values.

Fourth, the project did not include team strength variables. This is a major limitation because team quality is probably one of the strongest predictors of match outcomes. Variables such as club market value, Elo rating, UEFA ranking, squad value, or recent form could significantly improve the analysis.

Fifth, the machine learning models used only weather and travel-related variables. This was intentional because the project focused on external environmental and travel factors, but it limited model performance.

Sixth, the analysis treated all Champions League matches similarly. However, group-stage matches and knockout-stage matches may have different dynamics. Some teams may rotate players in less important group-stage matches, while knockout matches may be more intense and tactical.

Finally, some stadium and location information required manual completion. Although care was taken to make the reference table accurate, manually created reference data can still contain small errors.

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## 12. Future Work

Future work could improve this project in several ways.

First, team strength variables should be added. Including club market value, UEFA coefficient, Elo rating, squad value, or recent form would likely improve the ability to explain and predict match outcomes.

Second, travel distance could be measured more precisely. Instead of using representative home stadium coordinates, future work could use actual travel routes, flight distances, or travel time.

Third, weather data could be matched to kickoff time rather than using daily values. This would provide more accurate match-specific weather conditions.

Fourth, additional weather variables could be added, such as humidity, pressure, snowfall, apparent temperature, or pitch condition indicators.

Fifth, the dataset could be expanded beyond the UEFA Champions League. Including domestic league matches or other European competitions could increase the number of observations and provide more variation in travel patterns.

Sixth, future machine learning models could include interaction terms. For example, travel distance may matter more under certain weather conditions, or weather effects may differ depending on the country or stadium.

Finally, more advanced models such as gradient boosting, XGBoost, or models with team fixed effects could be tested. However, these models would be more meaningful if additional team-quality variables were included.

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## 13. Conclusion

This project investigated whether travel distance and weather conditions are associated with away-team performance in UEFA Champions League matches.

The project followed a complete data science workflow, including data collection, data cleaning, feature engineering, external data enrichment, exploratory analysis, hypothesis testing, and machine learning.

The exploratory analysis showed that away teams generally performed worse than home teams. Away teams won less often than they failed to win, and the average away goal difference was negative.

However, the hypothesis testing results did not provide statistically significant evidence that weather conditions or estimated travel distance strongly affected away-team goal difference. Temperature, precipitation, wind speed, and travel distance all showed weak or insignificant relationships with away performance.

The machine learning results supported this conclusion. Models trained only on weather and travel-related variables had limited predictive power for away wins. Models with higher accuracy mostly relied on predicting the majority class, while balanced models improved away-win detection but still produced modest results.

Overall, the project suggests that weather and travel distance may have some limited relationship with football performance, but they are not strong standalone predictors of away-team success in this dataset. More complete models would likely need to include team quality, tactical context, squad strength, recent form, and match importance.

The most important takeaway from this project is that external conditions such as weather and travel are interesting and measurable factors, but football performance is complex. Away-team success in the UEFA Champions League cannot be explained by environmental and travel-related variables alone.

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## 14. AI Usage Disclosure

This project was completed with limited AI assistance. AI was used mainly for organizing the notebook structure, improving written explanations, debugging Python code, and refining the presentation of the statistical and machine learning analyses.

AI support was also used to help structure the final report and improve the clarity of the discussion. However, the core project decisions were reviewed and decided by the student. These decisions include the research topic, dataset choice, enrichment strategy, feature selection, hypothesis testing design, machine learning model selection, interpretation of results, and final conclusions.



All code used in the final notebooks was executed, checked, adapted, and verified by the student. The outputs used in the report were generated from the student's own notebooks and reviewed before being included in the final interpretation.

AI was used as a supplementary support tool during the development of the project and not as a substitute for the student's own reasoning, analysis, and interpretation.